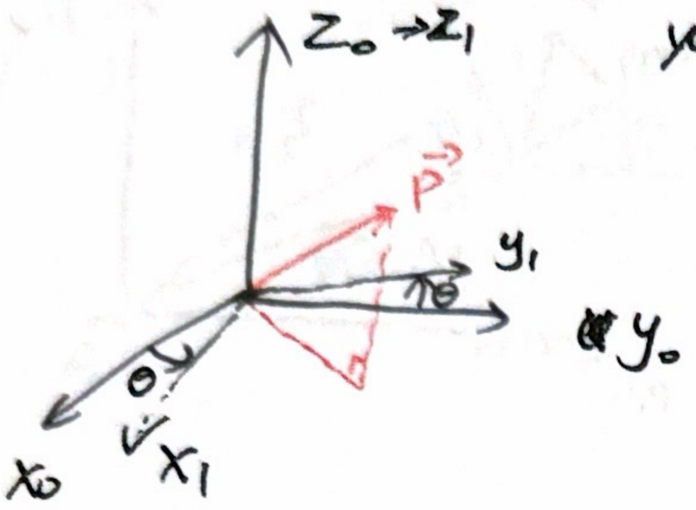
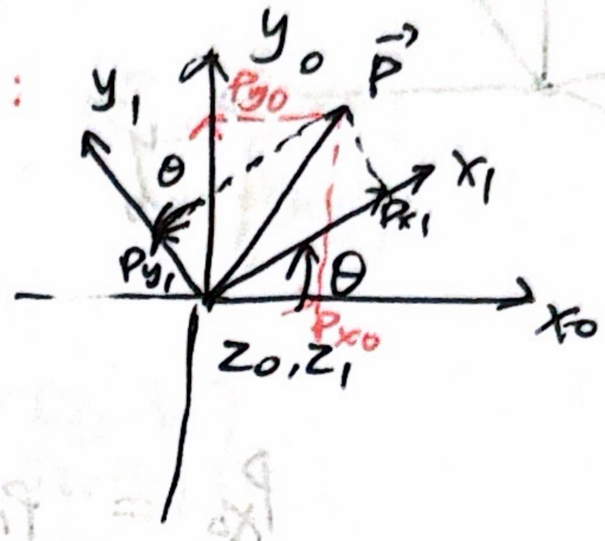
Rotational matrix:



yaw rotation:

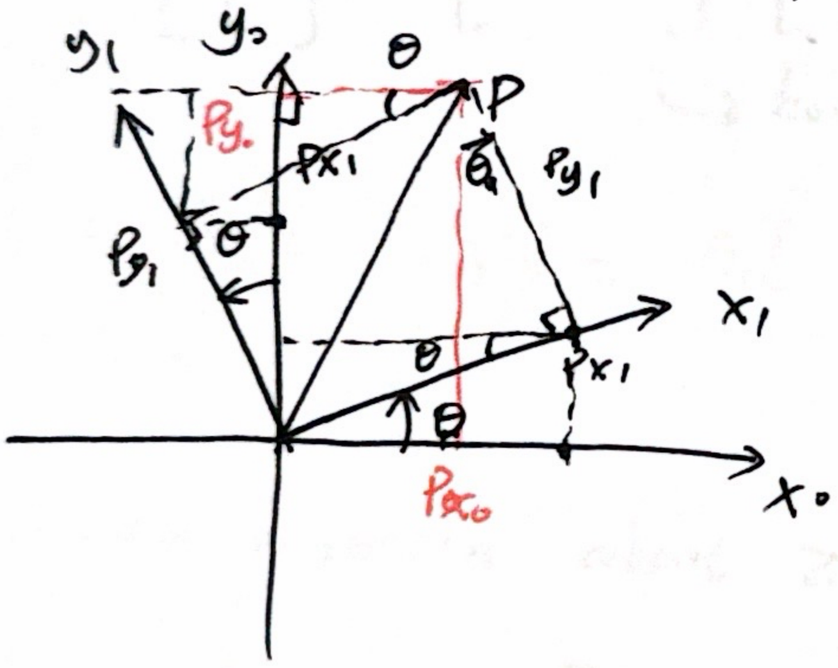
↳ project on x-y plane:



$$^1\vec{P} = \begin{bmatrix} P_{1x} \\ P_{1y} \end{bmatrix} \quad ; \quad ^0\vec{P} = \begin{bmatrix} P_{0x} \\ P_{0y} \end{bmatrix}.$$

find ^0_1R → to transform form 1 to 0 coordinate. such that.

$$^0\vec{P} = {}^0_1R \cdot {}^1\vec{P}$$

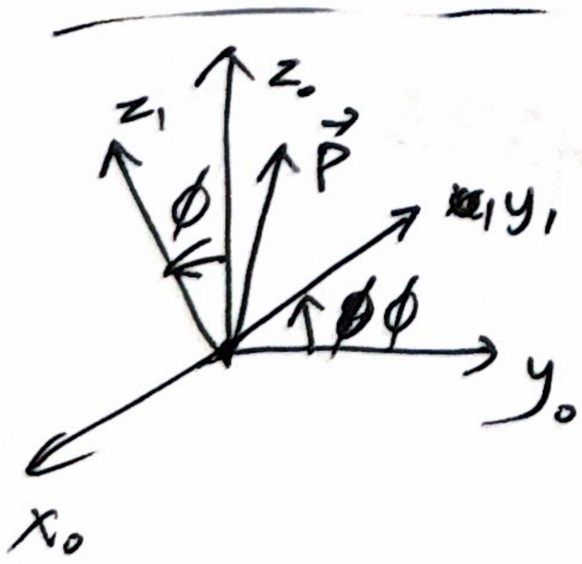


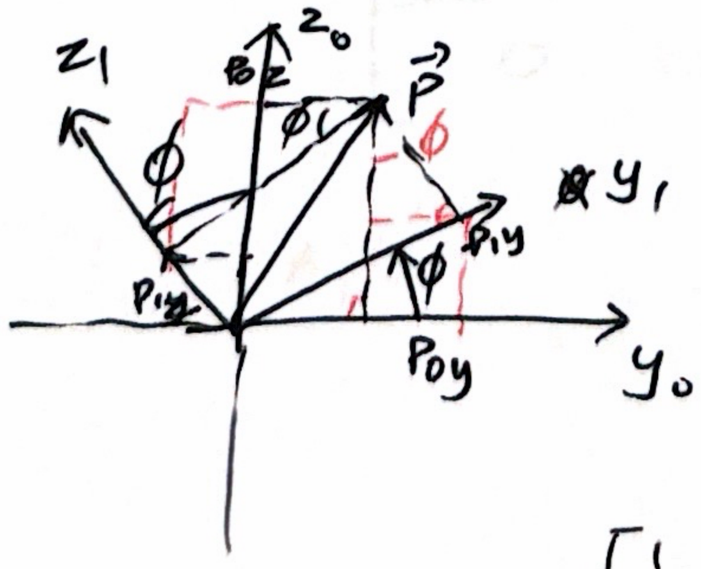
$$\begin{cases} P_{x0} = P_{x1} \cdot \cos\theta - P_{y1} \sin\theta . \\ P_{y0} = P_{y1} \cos\theta + P_{x1} \cdot \sin\theta \end{cases}$$

$$\begin{bmatrix} P_{x0} \\ P_{y0} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} P_{x1} \\ P_{y1} \end{bmatrix}$$

So

$$\begin{bmatrix} P_{x0} \\ P_{y0} \\ P_{z0} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} P_{x1} \\ P_{y1} \\ P_{z1} \end{bmatrix}$$ yaw. (ψ)





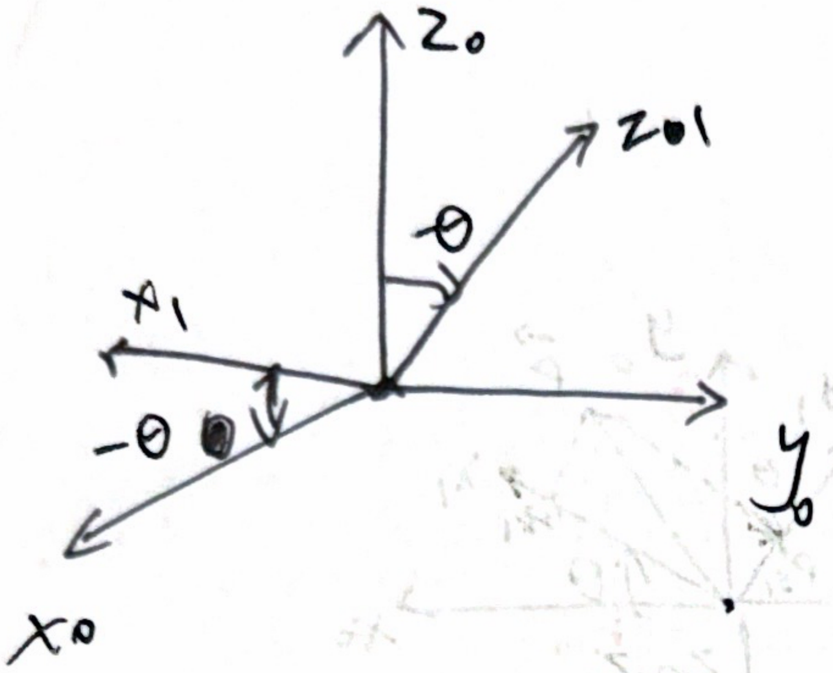
ϕ = roll angle.

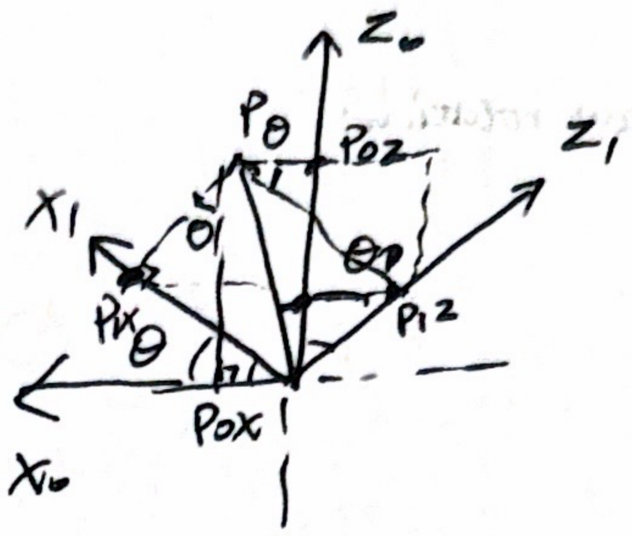
$$^1\vec{P} = \begin{bmatrix} P_{1y} \\ P_{1z} \end{bmatrix} \quad , \quad ^0\vec{P}_0 = \begin{bmatrix} P_{oy} \\ P_{oz} \end{bmatrix}$$

$$P_{oy} = P_{1y}\cos\phi - P_z \sin\phi$$

$$P_{oz} = P_{1z}\cos\phi + P_{1y}\sin\phi$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$



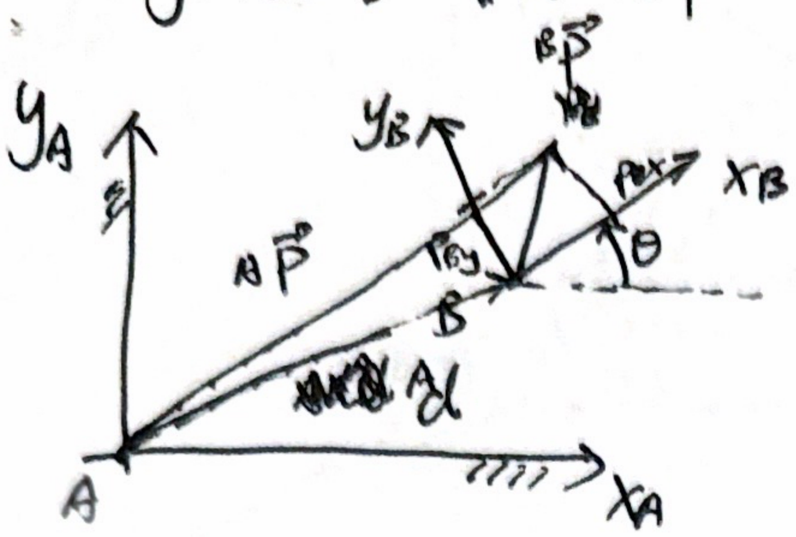


$$P_{X_0} = P_{1X}\cos\theta - P_{1Z}\sin\theta$$

$$P_{Z_0} = P_{1Z}\cos\theta + P_{1X}\sin\theta$$

$$\begin{bmatrix} \cos\theta & 0 & +\sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \quad \text{roll } (\theta).$$

Homogeneous Transformation Matrix:



$$^B\vec{P} = \begin{bmatrix} P_{Bx} \\ P_{By} \end{bmatrix}$$

given θ, $^A\vec{d}$, $^B\vec{P} = \begin{bmatrix} P_{Bx} \\ P_{By} \end{bmatrix}$

find $^A\vec{P} = \begin{bmatrix} P_{Ax} \\ P_{Ay} \end{bmatrix}$

find a matrix such that:

$$\begin{bmatrix} ^AP \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} \quad \end{bmatrix} \cdot \begin{bmatrix} ^BP \\ \vdots \\ 1 \end{bmatrix}$$

↳ homogeneous transformation matrix.

solution: $$^A\vec{P} = {}^A\vec{d} + {}^A_BR \cdot {}^B\vec{P}$$

↳ rotational matrix that transform from B to A.

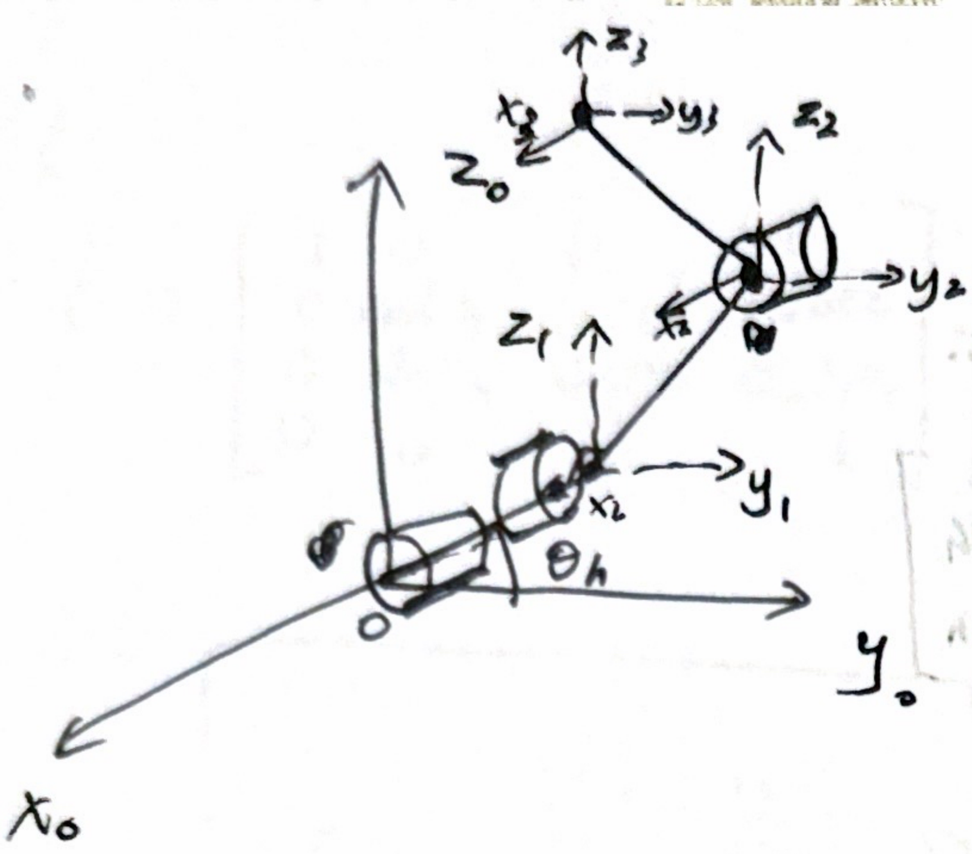
Since rotate about z:

$$^A_BR = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

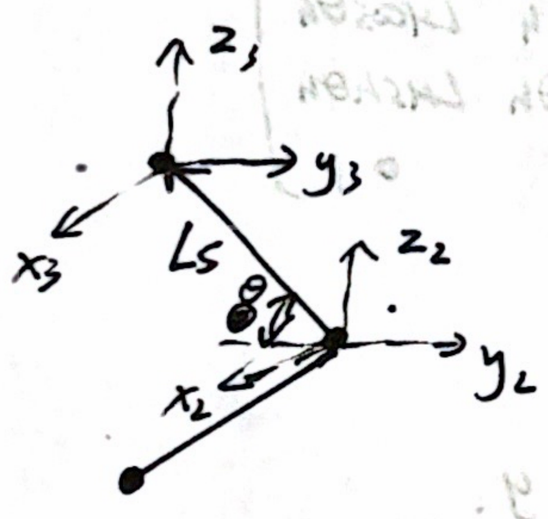
So $$^AP\,^A\vec{P} = {}^A\vec{d} + \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} P_{Bx} \\ P_{By} \\ P_{Bz} \end{bmatrix}.$$

$$= \begin{bmatrix} {}^A_BR & {}^Ad \\ 0_{2\times1} & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & {}^Ad_{xA} \\ \sin\theta & \cos\theta & d_{xy} \\ 0 & 0 & 1 \end{bmatrix}$$

homogeneous transform.



for shank :

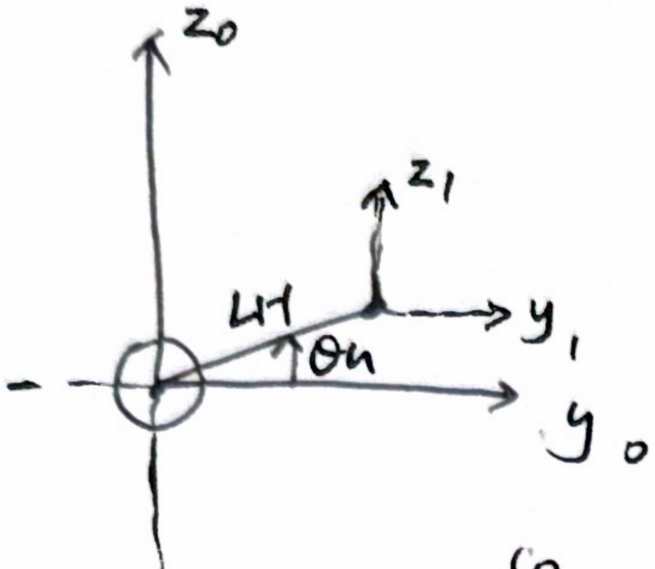


rotation above shank x.

$$\begin{cases} \ddot{L}s_x = \ddot{0} = 0, \\ Ls_y = -Ls\cos\theta \\ Ls_z = Ls\sin\theta. \end{cases}$$

$$\begin{bmatrix} 1 & 0 & 0 & L_{H}x_0 \\ 0 & \cos\theta_h & -\sin\theta_h & L_{H}y_0 \\ 0 & \sin\theta_h & \cos\theta_h & L_{H}z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta_t & 0 & \sin\theta_t & L_t x_1 \\ 0 & 1 & 0 & L_t y_1 \\ -\sin\theta_t & 0 & \cos\theta_t & L_t z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & Ls_{x_2} \\ 0 & \cos\theta_s & -\sin\theta_s & Ls_{y_2} \\ 0 & \sin\theta_s & \cos\theta_s & Ls_{z_2} \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

for hip:

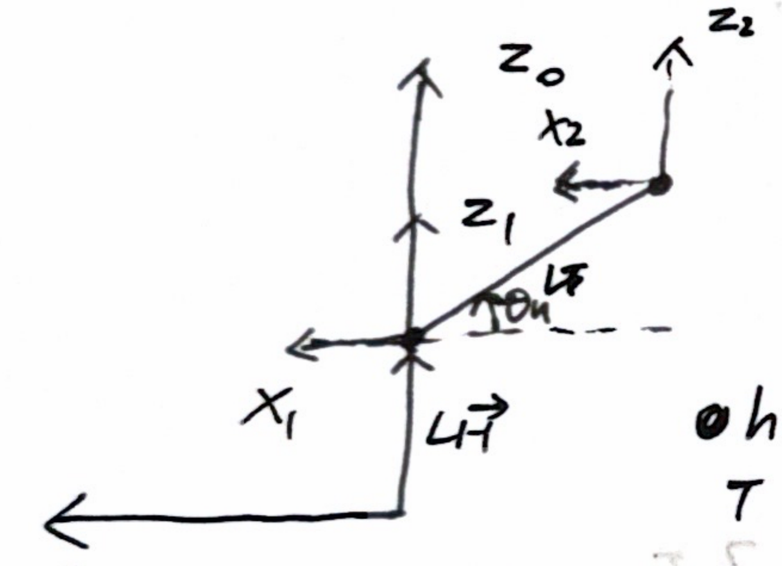


rotate about x.

$$\vec{L}_{H_0} = \begin{bmatrix} 0 \\ L_H \cos\theta_h \\ L_H \sin\theta_h \end{bmatrix}$$

$$\text{so } {}^0_h T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta_h & -\sin\theta_h & L_H\cos\theta_h \\ 0 & \sin\theta_h & \cos\theta_h & L_H\sin\theta_h \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

for thigh:

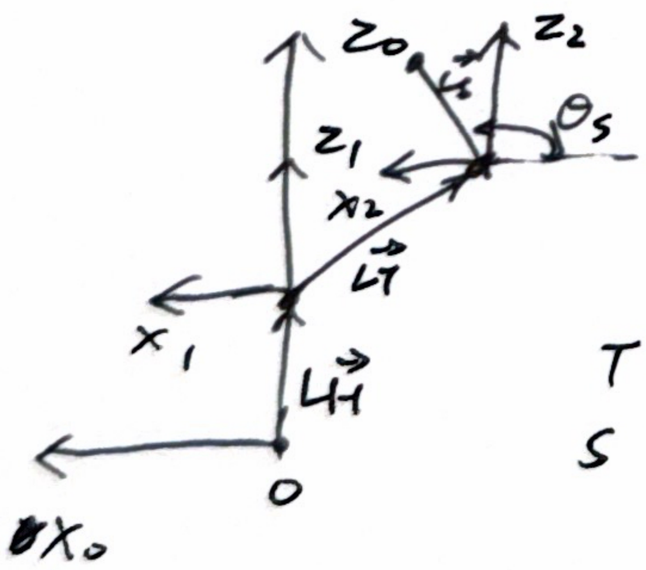


rotate about y.

$$\vec{L_{T_1}} = \begin{bmatrix} -L_T\cos\theta_t \\ 0 \\ L_T\sin\theta_t \end{bmatrix}$$

$${}^h_T T = \begin{bmatrix} \cos\theta_t & 0 & \sin\theta_t & -L_T\cos\theta_t \\ 0 & 1 & 0 & 0 \\ -\sin\theta_t & 0 & \cos\theta_t & L_T\sin\theta_t \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

for shank:



rotate about y.

$$\vec{L_S} = \begin{bmatrix} -L_S\cos\theta_s \\ 0 \\ L_S\sin\theta_s \end{bmatrix}$$

$${}^T_S T = \begin{bmatrix} \cos\theta_s & 0 & \sin\theta_s & -L_S\cos\theta_s \\ 0 & 1 & 0 & 0 \\ -\sin\theta_s & 0 & \cos\theta_s & L_S\sin\theta_s \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta_h & -\sin\theta_h & L_h\cos\theta_h \\ 0 & \sin\theta_h & \cos\theta_h & L_h\sin\theta_h \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta_t & 0 & \sin\theta_t & -L_t\cos\theta_t \\ 0 & 1 & 0 & 0 \\ -\sin\theta_t & 0 & \cos\theta_t & L_t\sin\theta_t \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\cos\theta_t \quad 0 \quad \sin\theta_t \quad -L_t\cos\theta_t$$

$$\cos\theta_t\cos\theta_h$$

$$\cos\theta_h\sin\theta_t - \sin\theta_h \quad \cos\theta_h\sin\theta_t - \sin\theta_h\cos\theta_t \quad -L_t\sin\theta_h\sin\theta_t + L_h\cos\theta_h$$

$$\cos\theta_h\cos\theta_t + \sin\theta_h\sin\theta_t \quad -\sin\theta_h \quad \cos\theta_h\sin\theta_t - \sin\theta_h\cos\theta_t \quad -L_t\sin\theta_h\sin\theta_t + L_h\cos\theta_h$$

$$\cos\theta_h\cos\theta_t \quad \cos\theta_h$$

$$\cos\theta_t \quad 0 \quad \sin\theta_t \quad -L_t\cos\theta_t$$

$$\sin\theta_h\sin\theta_t \quad \cos\theta_h \quad -\sin\theta_h\cos\theta_t \quad -L_t\sin\theta_t\sin\theta_h + L_h\cos\theta_h$$

$$-\cos\theta_h\sin\theta_t \quad \sin\theta_h \quad \cos\theta_h\cos\theta_t \quad +L_t\sin\theta_t\cos\theta_h + L_h\sin\theta_h.$$

$$0 \quad 0 \quad 0 \quad 1$$

So

$$T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta_h & -\sin\theta_h & L_h\cos\theta_h \\ 0 & \sin\theta_h & \cos\theta_h & L_h\sin\theta_h \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta_t & 0 & \sin\theta_t & -L_t\cos\theta_t \\ 0 & 1 & 0 & 0 \\ -\sin\theta_t & 0 & \cos\theta_t & L_t\sin\theta_t \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta_t & 0 & \sin\theta_t & -L_t\cos\theta_t \\ \sin\theta_h\sin\theta_t & \cos\theta_h & -\sin\theta_h\cos\theta_t & -L_t\sin\theta_t\sin\theta_h + L_h\cos\theta_h \\ -\cos\theta_h\sin\theta_t & \sin\theta_h & \cos\theta_h\cos\theta_t & +L_t\sin\theta_t\cos\theta_h + L_h\sin\theta_h \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta_s & 0 & \sin\theta_s & -L_s\cos\theta_s \\ 0 & 1 & 0 & 0 \\ -\sin\theta_s & 0 & \cos\theta_s & L_s\sin\theta_s \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot$$

$$\begin{bmatrix} \cos\theta_t\cos\theta_s - \sin\theta_t\sin\theta_s & 0 & \cos\theta_t\sin\theta_s + \sin\theta_t\cos\theta_t & -L_s\cos\theta_s\cos\theta_t + L_s\sin\theta_s\sin\theta_t - L_t\cos\theta_t \\ \sin\theta_h\sin\theta_t\cos\theta_s - \sin\theta_h\cos\theta_t\sin\theta_s & \cos\theta_h & \sin\theta_h\sin\theta_t\sin\theta_s - \sin\theta_h\cos\theta_t\cos\theta_s & -L_t\cos\theta_s\sin\theta_h\sin\theta_t - L_s\sin\theta_s\sin\theta_h\cos\theta_t \\ -\cos\theta_h\sin\theta_t\cos\theta_s - \cos\theta_h\cos\theta_t\sin\theta_s & \sin\theta_h & -\sin\theta_s\cos\theta_h\sin\theta_t + \cos\theta_s\cos\theta_h\cos\theta_t & -L_t\sin\theta_t\sin\theta_h + L_h\cos\theta_h \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$+L_s\cos\theta_s\cos\theta_h\sin\theta_t + L_s\sin\theta_s\cos\theta_h\cos\theta_t + L_t\sin\theta_t\cos\theta_h + L_h\sin\theta_h$$

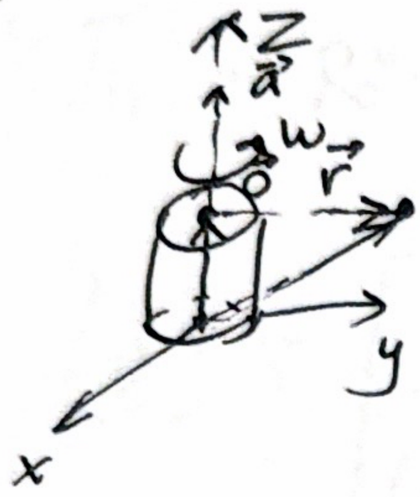
$$T = \begin{bmatrix} T & P \\ & \end{bmatrix}$$

FK.

since foot frame origin is at foot endpoint,

$$\text{so } {}^0\vec{P}_f = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow \text{forward k} = \begin{bmatrix} T_{14} \\ T_{24} \\ T_{34} \\ T_{44} \end{bmatrix}.$$

Jacobian:



$\vec{a}$: unit vector on rotation axis.

$\vec{O}$: a point on $\vec{a}$.

$\vec{P}$: end effector position.

$\vec{r} = \vec{P} - \vec{O}$

$\vec{\omega} = \vec{a} \cdot \dot{q}$. $\dot{q}$ = rotational speed of joint.

rigid body dynamic: $\dot{P} = \vec{\omega} \times \vec{r} = \left[\vec{a} \times (\vec{P} - \vec{O}) \right] \dot{q}$

$$J = \begin{bmatrix} \vec{a}_i \times (\vec{P}_i - \vec{O}_i) \\ \vec{a}_i \end{bmatrix} \cdot \quad \rightarrow \begin{bmatrix} \dot{P} \\ \vec{\omega} \end{bmatrix} = \begin{bmatrix} \vec{a}_i \times (P_i - \vec{O}_i) \\ \vec{a}_i \end{bmatrix} \cdot \dot{q}$$

$$\begin{cases} x = -L_s \cos\theta_s \cos\theta_t + L_s \sin\theta_s \sin\theta_t - L_t \cos\theta_t \\ y = -L_s \cos\theta_s \sin\theta_h \sin\theta_t - L_s \sin\theta_s \sin\theta_h \cos\theta_t - L_t \sin\theta_t \sin\theta_h + L_h \cos\theta_h \\ z = L_s \cos\theta_s \cos\theta_h \sin\theta_t + L_s \sin\theta_s \cos\theta_h \cos\theta_t + L_t \sin\theta_t \cos\theta_h + L_h \sin\theta_h \end{cases}$$

$$\begin{cases} x = -L_s C\theta_s C\theta_t + L_s S\theta_s S\theta_t - L_t C\theta_t \\ y = -L_s C\theta_s S\theta_h S\theta_t - L_s S\theta_s S\theta_h C\theta_t - L_t S\theta_t S\theta_h + L_h C\theta_h \\ z = L_s C\theta_s C\theta_h S\theta_t + L_s S\theta_s C\theta_h C\theta_t + L_t S\theta_t C\theta_h + L_h S\theta_h \end{cases}$$

$$\cos\alpha = \sin\alpha = \frac{1}{A}$$

$$A\cos\alpha = \quad \alpha = 45^\circ$$

$$A = \frac{1}{\sqrt{2}}$$

$$A\sin\alpha =$$

$$J_{11} = R_0^0 \cdot \begin{bmatrix}1\\0\\0\end{bmatrix} \times (d_3^0 - d_0^0)$$

d_2^0 → x, d_1^0 → y, → z

$$= \begin{bmatrix}1\\0\\0\end{bmatrix} \times \begin{bmatrix} -L_s C\theta_s C\theta_t + L_s S\theta_s S\theta_t - L_t C\theta_t \\ -L_s C\theta_s S\theta_h S\theta_t - L_s S\theta_s S\theta_h C\theta_t - L_t S\theta_t S\theta_h + L_h C\theta_h \\ L_s C\theta_s C\theta_h S\theta_t + L_s S\theta_s C\theta_h C\theta_t + L_t S\theta_t C\theta_h + L_h S\theta_h \end{bmatrix}$$

$$\begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ x & y & z \end{vmatrix} = i\begin{vmatrix} 0 & 0 \\ y & z \end{vmatrix} - j\begin{vmatrix} 1 & 0 \\ x & z \end{vmatrix} + k\begin{vmatrix} 1 & 0 \\ x & y \end{vmatrix}$$

$$= 0\hat{i} - j(z) + \hat{k}(y)$$

$$= \begin{bmatrix} 0 \\ -z \\ y \end{bmatrix} \rightarrow \text{col 1.}$$

hip rotates about leg frame x.

So $R_1^0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_h & -\sin\theta_h \\ 0 & \sin\theta_h & \cos\theta_h \end{bmatrix}$.

$$J_{12} = R_1^0 \begin{bmatrix}0\\1\\0\end{bmatrix} \times (d_3^0 - d_1^0)$$

$$= \begin{bmatrix} \cos\theta_t & 0 & \sin\theta_t \\ 0 & 1 & 0 \\ -\sin\theta_t & 0 & \cos\theta_t \end{bmatrix} \begin{bmatrix}0\\1\\0\end{bmatrix} \times (d_3^0 - d_1^0)$$

$$= \begin{bmatrix}0\\1\\0\end{bmatrix} \times \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \rightarrow \begin{vmatrix} i & j & k \\ 0 & 1 & 0 \\ x_1 & y_1 & z_1 \end{vmatrix} \rightarrow i\begin{vmatrix} 1 & 0 \\ y_1 & z_1 \end{vmatrix} - j\begin{vmatrix} 0 & 0 \\ x_1 & z_1 \end{vmatrix} + \hat{k}\begin{vmatrix} 0 & 1 \\ x_1 & y_1 \end{vmatrix}$$

$$= z_1\hat{i} - 0\hat{j} - x_1\hat{k}$$

$$= \begin{bmatrix} z_1 \\ 0 \\ -x_1 \end{bmatrix} \rightarrow \text{col 2}$$

$$z_1 = L_s C\theta_s C\theta_h S\theta_t + L_s S\theta_s C\theta_h C\theta_t + L_t S\theta_t C\theta_h.$$

$$x_1 = L_s C\theta_s C\theta_t + L_s S\theta_s S\theta_t - L_t C\theta_t.$$

So $J_{12} = R_1^0 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \times (d_3^0 - d_1^0)$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_4 & -\sin\theta_4 \\ 0 & \sin\theta_4 & \cos\theta_4 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \times (d_3^0 - d_1^0)$

$= \begin{bmatrix} 0 \\ \cos\theta_4 \\ \sin\theta_4 \end{bmatrix} \times \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \Rightarrow \begin{vmatrix} i & j & k \\ 0 & \cos\theta_4 & \sin\theta_4 \\ x_1 & y_1 & z_1 \end{vmatrix}$

$= i \begin{vmatrix} \cos\theta_4 & \sin\theta_4 \\ y_1 & z_1 \end{vmatrix} - j \begin{vmatrix} 0 & \sin\theta_4 \\ x_1 & z_1 \end{vmatrix} + k \begin{vmatrix} 0 & \cos\theta_4 \\ x_1 & y_1 \end{vmatrix}$

$= \begin{bmatrix} z_1\cos\theta_4 - y_1\sin\theta_4 \\ +x_1\sin\theta_4 \\ -x_1\cos\theta_4 \end{bmatrix} \rightarrow$ col 2.

$J_{13} = R_1^0 \cdot \begin{bmatrix} \cos\theta_5 & 0 & \sin\theta_5 \\ 0 & 1 & 0 \\ -\sin\theta_5 & 0 & \cos\theta_5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \times (d_3^0 - d_2^0)$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_4 & -\sin\theta_4 \\ 0 & \sin\theta_4 & \cos\theta_4 \end{bmatrix} \cdot \underbrace{\begin{bmatrix} \cos\theta_5 & 0 & \sin\theta_5 \\ 0 & 1 & 0 \\ -\sin\theta_5 & 0 & \cos\theta_5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}} \times (d_3^0 - d_2^0)$

$= \begin{bmatrix} 0 \\ \cos\theta_4 \\ \sin\theta_4 \end{bmatrix} \times (d_3^0 - d_2^0) = \begin{bmatrix} z_2\cos\theta_4 - y_2\sin\theta_4 \\ +x_2\sin\theta_4 \\ -x_2\cos\theta_4 \end{bmatrix} \rightarrow$ col 3.

$$J_{013} = R_2^0 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \times (d_3^0 - d_2^0)$$

$$= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \times \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} z_2 \\ 0 \\ -x_2 \end{bmatrix} \rightarrow \text{col 3.}$$

$$z_2 = L_5 C\theta_5 C\theta_4 S\theta_6 + L_5 S\theta_5 C\theta_4 C\theta_6$$

$$-x_2 = -L_5 C\theta_5 C\theta_6 + L_5 S\theta_5 S\theta_6$$

So $J(\theta_4, \theta_6, \theta_5) =$ [illegible]

$$R_2^0 = R_1^0 \cdot \begin{bmatrix} \cos\theta_6 & 0 & \sin\theta_6 \\ 0 & 1 & 0 \\ -\sin\theta_6 & 0 & \cos\theta_6 \end{bmatrix}$$

Summary :

forward kinematic :

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -L_s C\theta_s C\theta_t + L_s S\theta_s S\theta_t - L_t C\theta_t \\ -L_s C\theta_s S\theta_h S\theta_t - L_s S\theta_s S\theta_h C\theta_t - L_t S\theta_t S\theta_h + L_h C\theta_h \\ L_s C\theta_s C\theta_h S\theta_t + L_s S\theta_s C\theta_h C\theta_t + L_t S\theta_t C\theta_h + L_h S\theta_h \end{bmatrix}$$

$$\begin{cases} x_1 = x \\ y_1 = y - L_h C\theta_h \\ z_1 = z - L_h S\theta_h \end{cases} \quad \begin{cases} x_2 = x_1 - (-L_t C\theta_t) \\ y_2 = y_1 - (-L_t S\theta_t S\theta_h) \\ z_2 = z_2 - (L_t S\theta_t C\theta_h) \end{cases}$$

without angular velocity :

$$J(\theta_h, \theta_t, \theta_s) = \begin{bmatrix} R_0^0 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \times (d_3^0 - d_0^0) & R_1^0 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \times (d_3^0 - d_1^0) & R_{01} \cdot R_2^0 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \times (d_3^0 - d_2^0) \end{bmatrix}$$

$R_0^0 = I(3\times3)$, $d_3^0 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $d_1^0 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $d_2^0 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$.

$$R_1^0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_h & -\sin\theta_h \\ 0 & \sin\theta_h & \cos\theta_h \end{bmatrix}. \quad R_2^0 = \begin{bmatrix} \cos\theta_t & 0 & \sin\theta_t \\ 0 & 1 & 0 \\ -\sin\theta_t & 0 & \cos\theta_t \end{bmatrix}.$$

$$J = \begin{bmatrix} 0 & z_1 C\theta_h - y_1 S\theta_h & Z_2 C\theta_h - y_2 S\theta_h \\ -z & x_1 S\theta_h & x_2 \cdot S\theta_h \\ y & -x_1 C\theta_h & x_2 C\theta_h \end{bmatrix}.$$

$$x = -L_s \cos(\theta_s + \theta_t) - L_t \cos\theta_t$$

$$y = L_s \sin\theta_h \left(\cos\theta_s \sin\theta_t - \sin\theta_s \cos\theta_t\right) - L_t \sin\theta_t \sin\theta_h + L_h \cos\theta_h$$

$$Y = -L_s \sin\theta_h \sin(\theta_s + \theta_t) - L_t \sin\theta_t \sin\theta_h + L_h \cos\theta_h$$

$$z = L_s \sin\theta_h \cos\theta_h \left(\sin(\theta_s + \theta_t)\right) + L_t \sin\theta_t \cos\theta_h + L_h \sin\theta_h$$

$$y+z = L_s \sin(\theta_s+\theta_t)\left(\cos\theta_h - \sin\theta_h\right) + L_t\left(\sin\theta_t \cos\theta_h - \sin\theta_h \cos\theta_h\right) + L_h\left(\cos\theta_h + \sin\theta_h\right)$$

$$y+z = L_s \sin(\theta_s+\theta_t)\cos\left(\frac{\pi}{4}+\theta_h\right)\frac{\sqrt{2}}{2} + L_t \sin\theta_t \left(\cos\left(\theta_h + \frac{\pi}{4}\right)\frac{\sqrt{2}}{2}\right) + L_h \left(\sin\left(\theta_h + \frac{\pi}{4}\right)\frac{\sqrt{2}}{2}\right)$$

$$L_s = \frac{x + L_t \cos\theta_t}{-\cos(\theta_s+\theta_t)}$$

$$-\left(x + L_t\cos\theta_t\right)\cdot\tan(\theta_s+\theta_t)\left(\frac{\sqrt{2}}{2}\right)\left(\cos\left(\frac{\pi}{4}+\theta_h\right)\right.$$

$$-L_t \sin\theta_t \cdot \frac{\sqrt{2}}{2}\cos\left(\theta_h + \frac{\pi}{4}\right)$$

$$+ \frac{\sqrt{2}}{2} L_h \sin\left(\theta_h + \frac{\pi}{4}\right) \qquad = y+z$$

$A = l_t \sin\theta_t + l_s \sin(\theta_t + \theta_s)$, $C = l_t \cos\theta_t + l_s \cos(\theta_t + \theta_s)$.

$$\begin{cases} x = -C \\ y = l_h \cos\theta_h - A \sin\theta_h \\ z = l_h \sin\theta_h + A \cos\theta_h \end{cases}$$

$y = l_h C - As \rightarrow C = \frac{y + As}{l_h}$

$z = \frac{y + As}{l_h} A + l_h s$

$y^2 + z^2 = l_h^2 \cos^2\theta_h - 2 l_h A \cos\theta_h \sin\theta_h + A^2 \sin^2\theta_h$

$l_h^2 z = Ay + A^2 s + l_h^2 s$

$+ l_h^2 \sin^2\theta_h + 2 l_h A \cos\theta_h \sin\theta_h + A^2 \cos^2\theta_h$

$= Ay + (y^2 + z^2 - l_h^2)\, s + l_h^2 s$

$\downarrow$

$l_h^2 + A^2 = y^2 + z^2 \rightarrow l_h^2 = y^2 + z^2 - A^2$

$l_h^2 y = Ay + (y^2 + z^2)\, s$

$A = \pm\sqrt{y^2 + z^2 - l_h^2} = l_t \sin\theta_t + l_s \sin(\theta_t + \theta_s)$.

$C = -x$

$A^2 + C^2 = A^2 + x^2$

$= l_t^2 \sin^2\theta_t + 2 l_t \sin\theta_t l_s \sin(\theta_t + \theta_s) + l_s^2 \sin^2(\theta_t + \theta_s)$

$+ l_t^2 \cos^2\theta_t + 2 l_t l_s \cos\theta_t \cos(\theta_t + \theta_s) + l_s^2 \cos^2(\theta_t + \theta_s)$

$= l_t^2 + 2 l_s l_t (\sin\theta_t \sin(\theta_t + \theta_s) + \cos\theta_t \cos(\theta_t + \theta_s)) + l_s^2$

$= y^2 + z^2 - l_h^2 + x^2$

$\cos\theta_h = \frac{z - l_h \sin\theta}{A}$

$\cos\theta_h (l_h^2 + A^2) = (y^2 + z^2) \cos\theta_h$

$(l_h^2 + A^2) \cos^2\theta_h + (l_h^2 + A^2) \sin^2\theta_h = y^2 + z^2 = l_h^2 + A^2$,

$l_h^2 y = Ay + (y^2 + z^2)\, s$.

$$y = lhC - As \rightarrow s = \frac{lhC - y}{A}$$

$$z = AC + lhs \leftarrow C = \frac{y + As}{lh}$$

$$Az = AC + lh(lhC - y) = AC + lh^2C - ylh$$

$$A^2z = lA^2C + lh^2AC - lyAlh$$

$$(y^2+z^2)z = (y^2+z^2)AC \qquad ylh = AC + lh^2C - Az$$

$$Az = Ay + A^2s + lhs \qquad -yA + zlh = +A^2s + lhAs$$

$$-yA + zlh = (z^2+y^2-lh^2)s + lh^2s$$

$$lhy = lh^2C - Aslh. \qquad \begin{cases} yA = lhAC - A^2s \\ zlh = lhAC + lh^2As \end{cases}$$

$$Az = A^2C + lhAs$$

$$-yA + zlh = (z^2+y^2)s$$

$$lhy + Az = lh^2C + A^2C = (lh^2 + z^2 + y^2 - lh^2)C$$

$$lhy + Az = (z^2+y^2)C \qquad s = \frac{zlh - yA}{z^2+y^2}$$

$$C = \frac{lhy + Az}{z^2+y^2} \qquad \theta h = a\sin\left(\frac{zlh - yA}{z^2+y^2}\right)$$

$$\theta h = a\cos\left(\frac{lhy + Az}{z^2+y^2}\right)$$

$$\text{so} \quad \theta h = a\tan2\left(\frac{lhy + Az}{zlh - yA}, \frac{lhy + Az}{zlh + yA}\right)$$

$$C^2 + A^2 = l_t^2 + 2 l_s l_t \left(\sin\theta_t \sin(\theta_t + \theta_s) + \cos\theta_t \cos(\theta_t + \theta_s) \right) + l_s^2$$

$$= l_{\theta t}^2 + l_s^2 + 2 l_s l_t \cos(\theta_t - \theta_t - \theta_s)$$

$$C^2 + A^2 = l_t^2 + l_s^2 + 2 l_s l_t \cos(\theta_s)$$

$$\theta_s = a\cos\left(\frac{C^2 + A^2 - l_t^2 - l_s^2}{2 l_s l_t} \right)$$

$$A = l_t \sin\theta_t + l_s \sin\theta_t \cos\theta_s + l_s \sin\theta_s \cos\theta_t = (l_t + l_s \cos\theta_s) \sin\theta_t + l_s \sin\theta_s \cos\theta_t$$

$$C = (l_t + l_s \cos\theta_s) \cos\theta_t - l_s \sin\theta_s \sin\theta_t$$

$$\theta_t = a\tan 2(BA - DC, BC + DA)$$

$$B = l_t + l_s \cos(\theta_s), \quad D = l_s \sin(\theta_s)$$

So IK:

$$\theta_h = a\tan 2(z_h - y_A, l_h y + A z)$$

$$\theta_t = a\tan 2(BA - DC, BC + DA)$$

$$\theta_s = a\cos\left(\frac{C^2 + A^2 - l_t^2 - l_s^2}{2 l_s l_t} \right)$$

$$C = -x$$

$$A = \sqrt{y^2 + z^2 - l_h^2}$$

$$B = l_t + l_s \cos(\theta_s)$$

$$D = l_s \sin(\theta_s)$$